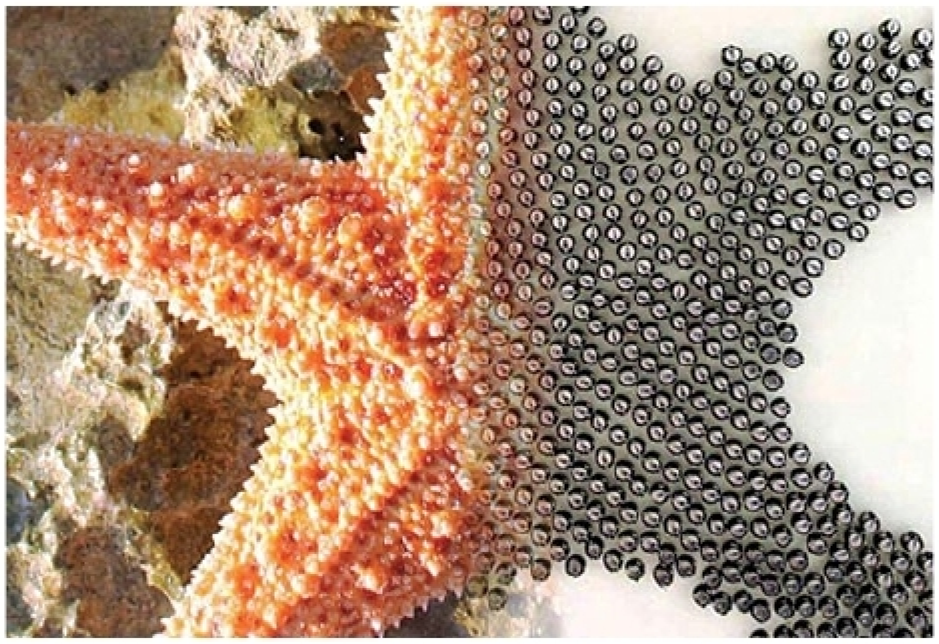

A 1024-robot swarm, capable of assembling itself into various shapes, developed at Harvard University (Credit: https://news.harvard.edu)

their efficiency. Interestingly, the field of micro-robotics became highly popular with the sci-fi movies *Fantastic Voyage* in 1966 and *Inner Space* in 1987, in which miniaturised submarine crews were injected inside the human body to perform non-invasive surgery.

The research and conceptual design of micro-robots commenced in the early 1970s as classified research for the US intelligence agencies for applications such as electronic intercept assignments. As basic miniaturisation support technologies were not fully developed at that time, progress was slow.

Present-day micro-bots were created in the last decade of the 20th century. One of the main challenges was to achieve their movement using the very limited power available from a small, lightweight battery source like a coin cell, or scavenging power from the surrounding environment in the form of vibration or light energy, or using biological motors like flagellated Serratia marcescen to draw chemical power from the surrounding fluid to actuate robotic devices. Progress in scanning electron microscopy (SEM) and scanning probe microscopy (SPM) in 1980s greatly helped the development of micro-robotics.

An NEMS having nanoscale transistors (left) and nano-scale mechanical components (right) (Credit: https://en.wikipedia.org/)

Initial studies in untethered micro-robotics included bacteria-inspired swimming propulsion, bacteria-propelled beads, steerable electrostatic crawling micro-bots, laser-powered micro-walkers, magnetic resonance imaging (MRI)-driven magnetic beads and magnetically-driven milli-scale nickel robots. Studies also progressed in actuation methods, such as helical propulsion, stick-slip crawling micro-bots, magneto tactic bacteria micro-bot swarms, optically-driven bubble micro-bots and micro-bots driven by a directed laser spot. The current approach is off-board (remote) actuation and control.

A few examples of the current notable developments are a 1024-robot swarm capable of assembling itself into various shapes developed at Harvard University; the program MicroFactory for Macro Products run by Defence Advanced Research Projects Agency (DARPA), an agency of Department of Defence for developing emerging technologies for the US military; and smart, highly-flexible, bio-compatible micro-bots designed in Zurich with the capability to swim through fluids, modify shape automatically when required, and pass through narrow blood vessels and the intricate human systems. There has been a surge in the research work to develop functionalised micro-bots, and the field is growing fast.